Circular-Cylinder Wakes with Lattice Boltzmann and Learned Prediction

Week 7 Extension Lecture Notes

M&I-ENG 690A – AI in Fluid Mechanics

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Central question. Which learned representation preserves convection, phase, strength, and shedding frequency on a held-out cylinder wake, rather than merely producing a visually smooth field?

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Contents

1 Position in the Course

The original course remains a six-week sequence: Weeks 1–4 establish numerical, kinetic, and surrogate-model foundations; the additive Week 4.1 develops classical POD–Galerkin and POD–DEIM for the cavity; Weeks 5–6 are the controlled research project. This cylinder module is a *Week 7 extension*. It changes the geometry and the dominant physics only after the original sequence is complete.

Learning checkpoint

By the end of Week 7, a student should be able to:

1. explain the physical transition from attached flow to steady recirculation and periodic vortex shedding;
2. derive the D2Q9 equilibrium distribution and BGK collision update;
3. connect Re, viscosity, relaxation time, and low-Mach restrictions;
4. measure drag, lift, recirculation length, and Strouhal number;
5. separate code verification, statistical convergence, grid convergence, domain sensitivity, and external validation;
6. diagnose a failed asymptotic/GCI sequence and specify the next refinement without deleting evidence or relaxing a gate;
7. design a complete-case Reynolds holdout without snapshot leakage; and
8. distinguish a low-error one-step forecast from a stable autonomous rollout, diagnose phase or amplitude failure, and explain when an explicit phase model is justified.

1.1 What is already executed

The release separates four claims that are often mixed together:

Claim	Evidence	Permitted wording
Wake regimes	five complete LBM cases with force histories and quality-gated frequency	physics-checked teaching evidence
Grid study	fixed-physics $N_D = 12, 18, 27$ sequence, statistical gates, fine-pair changes, and attempted Richardson/GCI analysis	statistical PASS; formal grid-independence FAIL retained
One-step neural forecast	four true prior fields, a complete historically held-out Reynolds case, and non-neural extrapolation baselines	low-error one-step interpolation
Autonomous rollout	recursive feedback for 50 steps with phase/amplitude diagnostics	audited limitation; not a passed claim
Phase-stable rollout	four true initial fields followed by 277 predictions with no future CFD input	passed only for the saturated, bracketed periodic regime

This distinction is central to the scientific method of the course. A visually smooth animation cannot promote a failed rollout into a validated result.

2 Cylinder-Wake Physics

For a cylinder of diameter D in a uniform stream U_∞, the governing nondimensional parameter is

$$\mathrm{Re}_D = \frac{U_\infty D}{\nu}. \tag{1}$$

At very low Re_D, the flow is attached and viscous. A symmetric recirculation bubble develops as inertia grows. The steady two-dimensional wake loses stability near $\mathrm{Re}_D \approx 47$, producing the alternating Bénard–von Kármán vortex street. The precise measured transition and forces depend on blockage, domain extent, discretization, and boundary treatment.

The relevant observables are

$$C_D = \frac{F_D}{\frac{1}{2}\rho U_\infty^2 D}, \qquad C_L = \frac{F_L}{\frac{1}{2}\rho U_\infty^2 D}, \tag{2}$$

$$\mathrm{St} = \frac{f_s D}{U_\infty}, \qquad \omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}. \tag{3}$$

Periodic lift is often the cleanest signal for estimating shedding frequency f_s. A credible field prediction should reproduce not only pixelwise vorticity but also the spectrum and phase of C_L and the downstream decay of enstrophy.

3 From Boltzmann Kinetics to D2Q9 LBM

The lattice Boltzmann method evolves discrete populations $f_i(\boldsymbol{x}, t)$ associated with lattice velocities $\boldsymbol{c}_i$. Density and momentum are moments:

$$\rho = \sum_i f_i, \qquad \rho\boldsymbol{u} = \sum_i f_i \boldsymbol{c}_i. \tag{4}$$

For D2Q9,

$$\boldsymbol{c}_i \in \{(0,0), (\pm 1, 0), (0, \pm 1), (\pm 1, \pm 1)\}, \tag{5}$$

with weights $w_0 = 4/9$, $w_{1\ldots4} = 1/9$, and $w_{5\ldots8} = 1/36$. The second-order isothermal equilibrium is

$$f_i^{eq} = w_i \rho \left[1 + \frac{\boldsymbol{c}_i \cdot \boldsymbol{u}}{c_s^2} + \frac{(\boldsymbol{c}_i \cdot \boldsymbol{u})^2}{2c_s^4} - \frac{\boldsymbol{u} \cdot \boldsymbol{u}}{2c_s^2}\right], \qquad c_s^2 = \frac{1}{3}. \tag{6}$$

3.1 Collision and streaming

The single-relaxation-time BGK update is

$$f_i^*(\boldsymbol{x}, t) = f_i(\boldsymbol{x}, t) - \frac{1}{\tau}\left[f_i(\boldsymbol{x}, t) - f_i^{eq}(\boldsymbol{x}, t)\right], \tag{7}$$

followed by streaming,

$$f_i(\boldsymbol{x} + \boldsymbol{c}_i \Delta t, t + \Delta t) = f_i^*(\boldsymbol{x}, t). \tag{8}$$

In lattice units,

$$\nu = c_s^2\left(\tau - \frac{1}{2}\right), \qquad \tau = \frac{1}{2} + \frac{U_\infty D}{c_s^2 \mathrm{Re}_D}. \tag{9}$$

Thus increasing Re_D at fixed U_∞ and D drives τ toward $1/2$, where numerical robustness becomes difficult. The notebook provides both transparent BGK and a more robust two-relaxation-time (TRT) option.

3.2 The complete algorithm in one loop

One LBM time step is compact, but its order matters:

1. recover ρ and $\boldsymbol{u}$ from the nine populations;
2. impose zero macroscopic velocity on solid nodes for diagnostics;
3. form $f_i^{eq}(\rho, \boldsymbol{u})$;
4. collide with BGK or the default TRT operator;
5. stream every post-collision population along $\boldsymbol{c}_i$;
6. reflect fluid–solid links with Bouzidi interpolation and accumulate momentum exchange for lift and drag; and
7. update the Zou–He inlet and convective outlet, then record mass, force, and any requested fields.

Initialization uses $f_i = f_i^{eq}$ and a half-cosine acceleration. Repeating the loop advances populations, not the Navier–Stokes variables directly; $(\rho, \boldsymbol{u}, p)$ are moments reconstructed from f_i. This distinction is why a valid restart must retain all nine populations and the outlet memory.

3.3 TRT collision and the magic parameter

Let $\bar{i}$ denote the direction opposite to i. Split the populations and their equilibria into symmetric and antisymmetric parts,

$$f_i^{\pm} = \frac{1}{2}(f_i \pm f_{\bar{i}}), \qquad f_i^{eq,\pm} = \frac{1}{2}(f_i^{eq} \pm f_{\bar{i}}^{eq}). \tag{10}$$

The TRT collision is

$$f_i^{*} = f_i - \omega^{+}\left(f_i^{+} - f_i^{eq,+}\right) - \omega^{-}\left(f_i^{-} - f_i^{eq,-}\right), \qquad \omega^{\pm} = \frac{1}{\tau^{\pm}}. \tag{11}$$

The viscous relaxation time is $\tau^{+} = \tau$ and therefore fixes ν. FlowMLLab selects τ^{-} through

$$\Lambda = \left(\tau^{+} - \frac{1}{2}\right)\left(\tau^{-} - \frac{1}{2}\right) = \frac{3}{16}, \qquad \tau^{-} = \frac{1}{2} + \frac{\Lambda}{\tau^{+} - 1/2}. \tag{12}$$

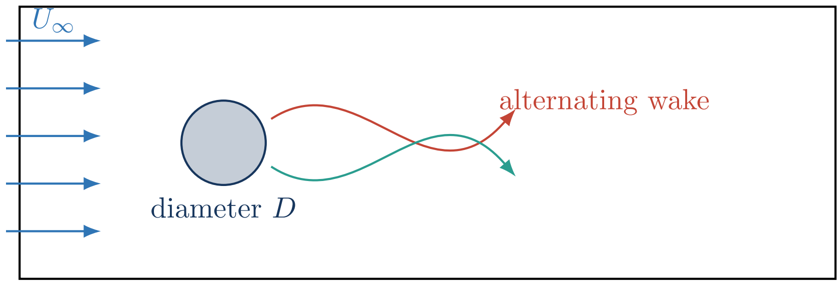
This does not change the Navier–Stokes viscosity; it controls the relaxation of odd non-equilibrium moments and improves wall-location robustness at small viscosity. The value $3/16$ is a documented numerical choice, not a fitted physical constant.

Important distinction

LBM is weakly compressible. Keep $\mathrm{Ma} = U_{\infty}/c_s$ small and monitor density drift. A correct nominal Reynolds number does not compensate for excessive Mach number, inadequate resolution, or boundary contamination.

4 Geometry and Boundary Conditions

The computational domain must provide sufficient upstream distance, downstream wake length, and transverse clearance. The cylinder uses a circular solid mask with no-slip bounce-back. The inlet prescribes velocity; the outlet must minimize reflected disturbances; upper and lower boundaries must be consistent with the intended blockage model.



The default wall is *not* plain staircase bounce-back. The binary mask identifies solid nodes, but every fluid-to-solid link intersects the analytical circle. If $q \in (0,1]$ is the fraction of the link from a fluid node $\boldsymbol{x}_f$ to that intersection, the Bouzidi update implemented here is

$$f_{\bar{i}}(\boldsymbol{x}_f, t+1) = 2q f_i^*(\boldsymbol{x}_f, t) + (1-2q) f_i^*(\boldsymbol{x}_f - \boldsymbol{c}_i, t), \qquad q < \frac{1}{2}, \tag{13}$$

$$f_{\bar{i}}(\boldsymbol{x}_f, t+1) = \frac{1}{2q} f_i^*(\boldsymbol{x}_f, t) + \frac{2q-1}{2q} f_{\bar{i}}^*(\boldsymbol{x}_f, t), \qquad q \geq \frac{1}{2}. \tag{14}$$

Halfway bounce-back remains available only as a controlled comparison. Finite diameter resolution still produces error, so an analytical link intersection does not remove the need for grid refinement.

4.1 Inlet, outlet, and transverse interpretation

At the west boundary, Zou–He reconstruction for $\boldsymbol{u} = (U_\infty, 0)$ gives

$$\rho = \frac{f_0 + f_2 + f_4 + 2(f_3 + f_6 + f_7)}{1 - U_\infty}, \tag{15}$$

$$f_1 = f_3 + \frac{2}{3}\rho U_\infty, \tag{16}$$

$$f_5 = f_7 + \frac{1}{2}(f_4 - f_2) + \frac{1}{6}\rho U_\infty, \tag{17}$$

$$f_8 = f_6 + \frac{1}{2}(f_2 - f_4) + \frac{1}{6}\rho U_\infty. \tag{18}$$

The east boundary uses a first-order convective distribution condition. Its implicit discrete form is

$$f_i^{n+1}(x_o) = \frac{f_i^n(x_o) + c_o f_i^{n+1}(x_o - \Delta x)}{1 + c_o}, \tag{19}$$

with $c_o = U_\infty$ unless explicitly changed. The top and bottom are periodic. Consequently, the mathematical problem is an infinite transverse array of cylinders with pitch L_y, not a literally isolated body. The blockage $\beta = D/L_y$ must therefore be reported; the retained profile uses $\beta \leq 0.125$.

4.2 Smooth startup and restart

An impulsive start excites weakly compressible acoustic modes. FlowMLLab supports a half-cosine startup from rest: the far-field populations are accelerated with a density-preserving equilibrium difference while inlet speed and convective-outlet memory follow the same ramp. For the short classroom path, the more reliable solution is a retained population checkpoint: all nine f_i fields and the convective-outlet memory are restored, and the student continues an already saturated wake. Restarting only from (u, v, p) is insufficient because it discards non-equilibrium moments.

5 Force and Frequency Extraction

The momentum-exchange method estimates the force transferred during population reflection. For each fluid-solid link, the momentum transferred to a stationary wall is the sum of outgoing and reflected link momentum,

$$\boldsymbol{F} = \sum_{\text{fluid–solid links}} \boldsymbol{c}_i \left[f_i^*(\boldsymbol{x}_f, t) + f_{\bar{i}}(\boldsymbol{x}_f, t+1) \right]. \tag{20}$$

The solver block-averages this lattice-step force before forming C_D and C_L. After discarding the transient, compute time-averaged drag and RMS lift.

5.1 A quality-gated Strouhal estimate

Let $t^* = tU_\infty/D$; frequency in t^* is directly St. Searching the entire FFT is unsafe because an impulsive start can excite a transverse acoustic period L_y/c_s. The retained estimator therefore requires all of the following:

1. the peak lies inside $0.05 < \mathrm{St} < 0.5$;
2. the retained window contains at least eight periods of that peak;
3. lift RMS in the second half differs by no more than 25% from the first;
4. the in-band peak exceeds the median in-band spectral background; and
5. a non-negligible fraction of total signal energy lies in the physical band.

If any gate fails, the reported result is `NaN` with a reason such as `fewer_than_minimum_cycles`; the code does not print a plausible-looking frequency.

5.2 Mean recirculation length

For an unsteady wake, the literature recirculation length is obtained from the post-transient time-mean streamwise velocity, not the last instantaneous field. On the wake centreline, let x_0 be the first downstream zero of $\overline{u}(x, 0)$ after the cylinder rear $x_r = x_c + D/2$. Then

$$L_r = x_0 - x_r, \qquad \frac{L_r}{D} = \frac{x_0 - x_r}{D}. \tag{21}$$

FlowMLLab retains both the time-mean value used for comparison and the instantaneous last-field value for teaching.

For a shedding case, evidence should include:

1. a sustained $C_L(t^*)$ oscillation after transient removal;
2. a narrow spectral peak and a stable St estimate across adjacent windows;
3. alternating signed vorticity in the wake;

4. mean C_D and St compared with appropriate two-dimensional references; and
5. sensitivity to grid, domain size, time step, and sampling window.

Evidence gate

The quick retained cases (Re $= 5, 20, 40, 100, 180$) are teaching evidence. They demonstrate expected regimes and diagnostics, but the qualification path—not a single attractive contour—is the basis for a quantitative claim.

6 Solution Verification: Grid Independence

Grid refinement must change one numerical scale while holding the physical problem fixed. For the executed Re $= 100$ study, the initial cylinder resolutions were

$$N_D = \frac{D}{\Delta x} = 12, 18, 27, \qquad r = 1.5. \tag{22}$$

All grids use Ma $= 0.0866$, a $20D \times 8D$ domain, cylinder centre $5D$ from the inlet, TRT collision, Bouzidi reflection, the same deterministic seed, $100D/U_\infty$ total time, and statistics after $45D/U_\infty$. Thus a change in an output can be attributed to spatial resolution rather than a simultaneous change in geometry, boundary placement, or sampling duration. With this constant-Mach acoustic scaling, $\tau = 1/2 + 3U_\infty N_D/\mathrm{Re}$ changes algebraically from one grid to the next to keep Re fixed. It is reported for every grid and is never tuned to improve an output.

The fine-pair changes are below their declared tolerances, but the successive differences did not decrease consistently; hence no positive asymptotic order or GCI can be accepted. That failure is retained. The next declared continuation is $N_D = 40$ with unchanged physics and gates, but it is not part of the completed evidence. For the executed constant-ratio sequence,

$$p = \frac{\ln |(\phi_1 - \phi_2)/(\phi_2 - \phi_3)|}{\ln r}, \tag{23}$$

$$\phi_{h\to 0} = \phi_3 + \frac{\phi_3 - \phi_2}{r^p - 1}, \tag{24}$$

$$\mathrm{GCI}_{\mathrm{fine}} = 1.25 \frac{|\phi_3 - \phi_2|/|\phi_3|}{r^p - 1} \times 100\%. \tag{25}$$

A GCI is reported only for a monotone asymptotic sequence. The predeclared course gates require medium-to-fine changes below 3%, 2%, and 5% for $\overline{C}_D$, St, and L_r/D, respectively; corresponding fine-grid GCI limits are 5%, 3%, and 8%. Every grid must also pass density, tail-drag, retained-cycle, and lift-stationarity gates.

N_D	τ	$\overline{C}_D$	St	L_r/D	statistical gate
12	0.5180	1.57220	0.181504	1.42222	PASS
18	0.5270	1.55524	0.181876	1.41770	PASS
27	0.5405	1.52262	0.182812	1.43726	PASS

The $18 \to 27$ changes are 2.142%, 0.512%, and 1.361% for $\overline{C}_D$, St, and L_r/D. They pass the practical sensitivity limits, but practical insensitivity is not a substitute for a positive-order asymptotic sequence; the formal decision remains FAIL.

Important distinction

Grid independence is necessary but not sufficient for validation. A solution may converge to the wrong answer because the transverse domain is narrow, the outlet is too close, Ma is too large, or the boundary model is biased. After this study, vary domain size at fixed fine resolution and only then compare $\overline{C}_D$, St, and L_r/D with independent CFD/DNS data.

The archived CNN and phase-decoder experiments remain tied to the low-cost $N_D = 12$ CFD dataset. The failed formal grid study does not retroactively upgrade those labels. A future qualified dataset must be regenerated only after grid and domain gates pass, and the learned models must then be retrained.

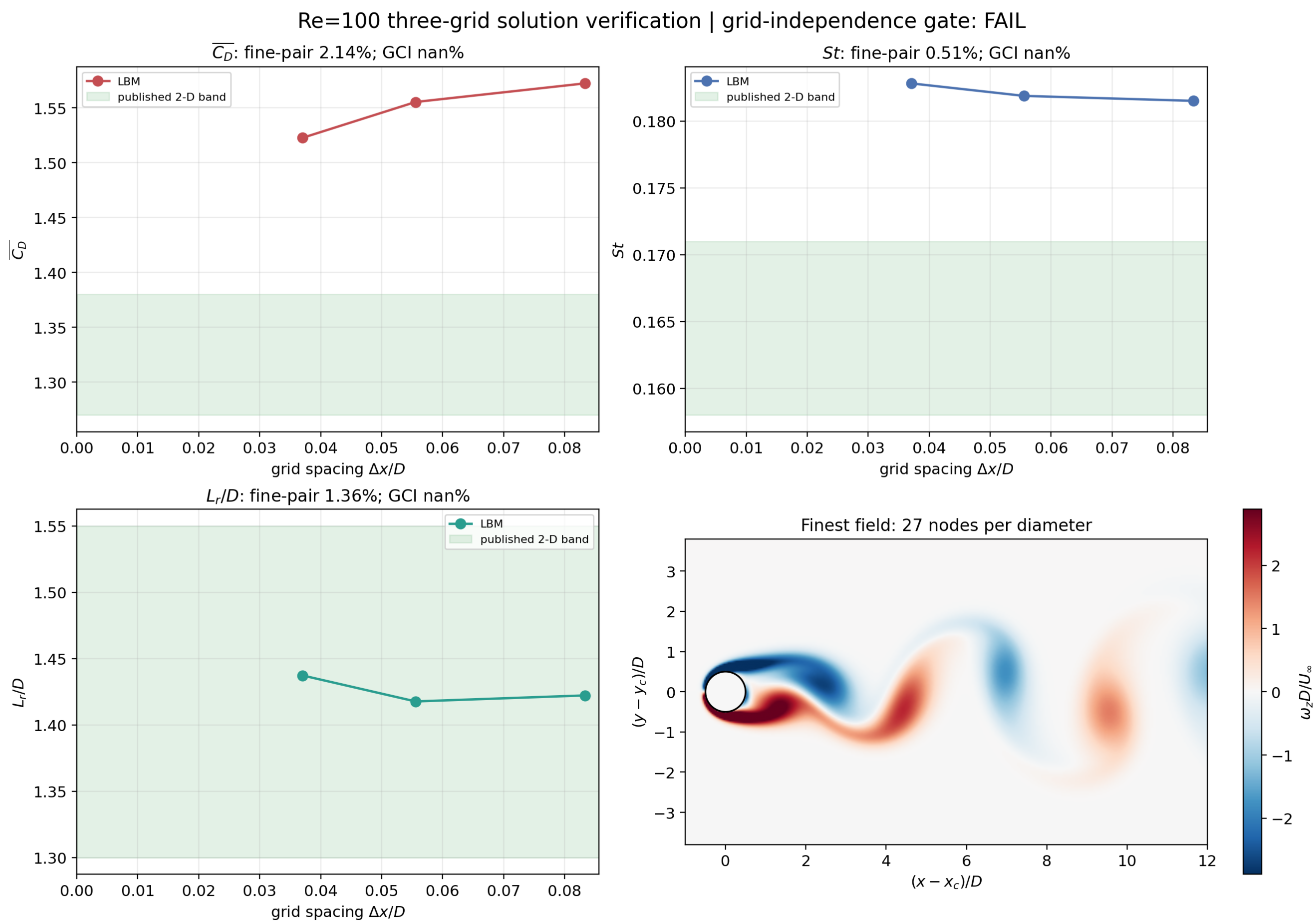


Figure 1: Executed three-grid solution verification at Re = 100. All fine-pair changes pass their declared sensitivity limits, but the formal asymptotic/GCI gate fails and is retained. The first three panels show the computed outputs, published comparison band, fine-pair change, and GCI. The finest vorticity panel uses equal physical axis scales, so the analytical cylinder remains circular.

6.1 The CFD-to-ML firewall

The CFD is qualified before a network is trained. This ordering prevents a small surrogate error from being mistaken for physical accuracy:

1. freeze the physical problem, nondimensional numbers, geometry, boundaries, solver, outputs, and acceptance gates;

2. verify implementation invariants, mass behaviour, no-slip, force signs, restart consistency, and statistical stationarity without loading an ML model;
3. run spatial-grid, domain/blockage, Mach/time-step, and sampling-window sensitivity studies one factor at a time;
4. compare only the qualified CFD outputs with independent CFD/DNS data and record discrepancies, provenance, configuration, and hashes;
5. freeze a versioned CFD dataset; only then create complete-Reynolds development, validation, and test splits for learning; and
6. treat ML-versus-CFD error as a separate conditional quantity. If the CFD protocol changes, create a new dataset version and retrain; never tune the CFD solver against neural predictions.

In this release, Step 3 has not passed its formal spatial-grid gate. Therefore the $N_D = 12$ neural labels are retained as classroom data, not promoted to a grid- and domain-qualified reference solution.

7 Executed LBM Evidence

Learning checkpoint

The short notebook path loads this versioned evidence and optionally continues the saturated $Re = 100$ population checkpoint. It does not cold-start a 4,500-step trajectory and pretend that a decaying symmetric wake is a developed vortex street.

8 Learning an Unsteady Wake

An unsteady wake is a transport problem. Vortices translate, deform, and decay. A low-rank linear combination of global modes can smear compact structures when phase is imperfect. Pixelwise mean-square error also rewards smooth predictions, so a network can reduce loss while destroying enstrophy downstream.

8.1 Leakage-free split

Snapshots from one trajectory are strongly correlated. Randomly distributing them among training and test sets is leakage. The Week-7 protocol holds out an entire Reynolds-number case. Model selection uses development cases only; the unseen case is opened once after architecture and hyperparameters are frozen.

8.2 Four-frame multi-scale CNN

Let four consecutive input fields be

$$\mathcal{X}_n = \left[\{u, v, p\}_{n-3}, \{u, v, p\}_{n-2}, \{u, v, p\}_{n-1}, \{u, v, p\}_n, \mathrm{Re}, \chi_f\right]. \tag{26}$$

A convolutional model predicts $\{\widehat{u}, \widehat{v}, \widehat{p}\}_{n+1}$; vorticity is derived from the predicted velocity. Multiple spatial scales help represent both the near-cylinder shear layers and larger downstream vortices. The fluid mask χ_f enforces exact zero velocity inside the analytical cylinder.

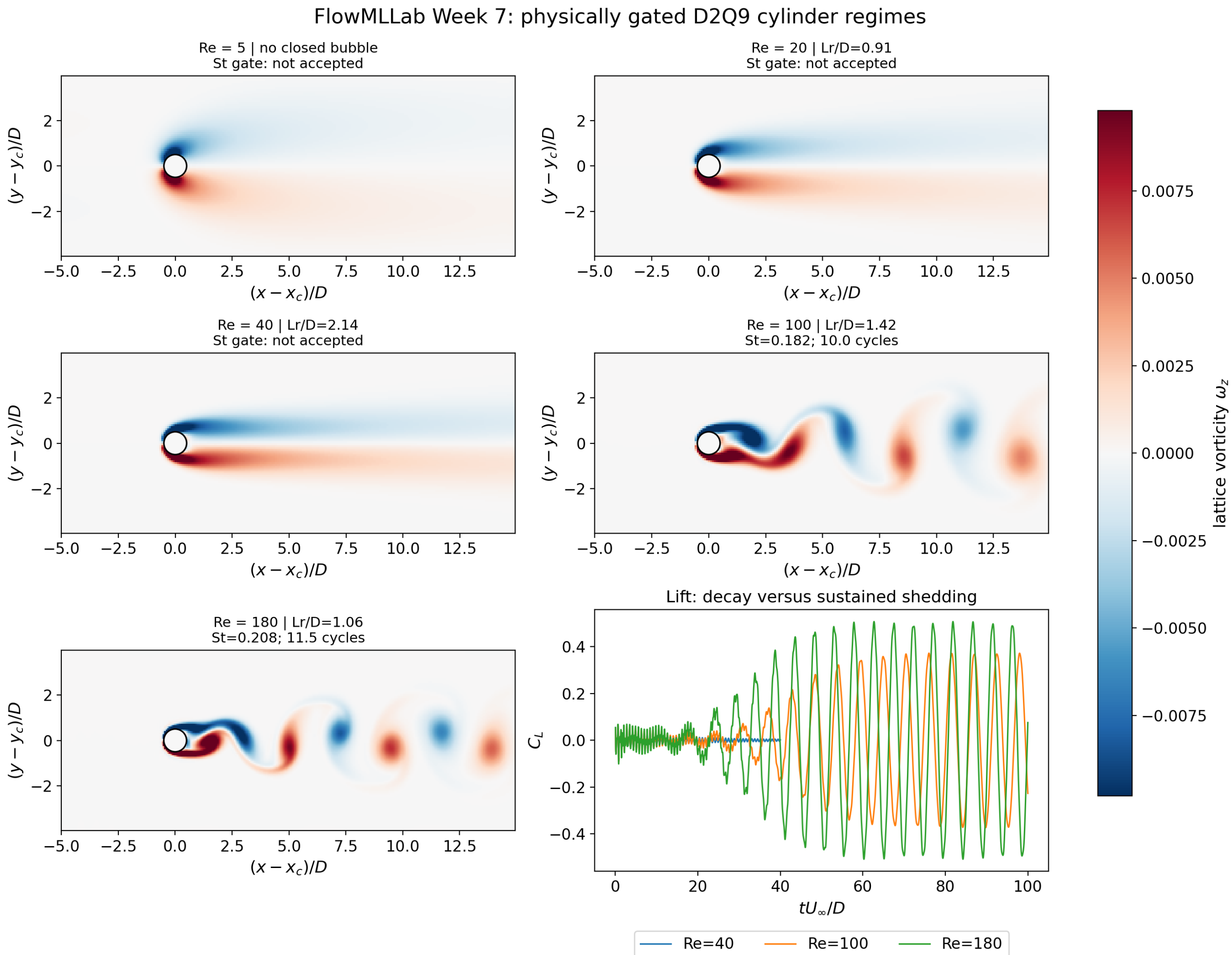


Figure 2: Retained D2Q9 regime evidence. The contour, force history, frequency quality flags, and time-mean recirculation length are interpreted together.

8.3 Matched baselines

Every forecast must be compared with methods that receive the same four frames. Persistence uses $\widehat{\boldsymbol{q}}_{n+1} = \boldsymbol{q}_n$. Polynomial extrapolation fits each field degree of freedom over $t = n-3, \ldots, n$:

$$\widehat{\boldsymbol{q}}^{(m)}_{n+h} = \sum_{k=0}^{m} \boldsymbol{a}_k (n+h)^k, \qquad m = 1, 2, 3. \tag{27}$$

These baselines are cheap and surprisingly strong for a short, smooth horizon. The public comparison therefore reports the best declared polynomial baseline, not persistence alone.

8.4 One step is not rollout

Teacher forcing supplies four true LBM frames for every next-frame prediction. Autonomous rollout supplies true frames only initially and recursively feeds the CNN output back. The input distribution then drifts, exposing accumulated phase and amplitude error. These are different experiments and must be labelled separately.

Useful evaluation quantities include

$$\epsilon_{L_2}(t) = \frac{\|\widehat{\omega}(t) - \omega(t)\|_2}{\|\omega(t)\|_2}, \tag{28}$$

$$E_\omega(x_1, x_2) = \frac{1}{2}\int_{x_1}^{x_2} \int \omega^2 \, dy \, dx, \tag{29}$$

$$\epsilon_\phi(t) = \mathrm{wrap}\left(\widehat{\phi}(t) - \phi(t)\right). \tag{30}$$

Report the downstream enstrophy ratio, spectral peak, and phase error together with global L_2 error.

9 Executed Neural Evidence

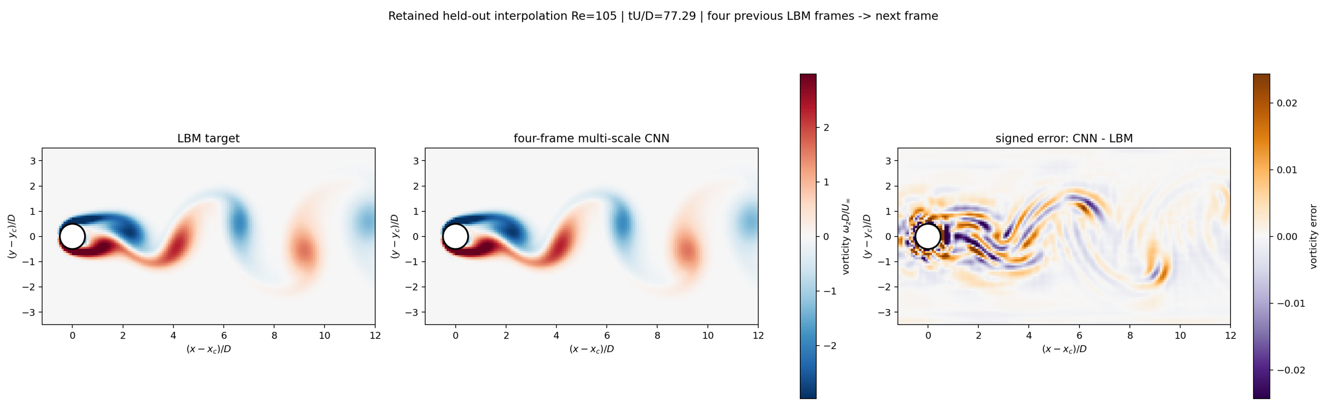


Figure 3: Retained $Re = 105$ one-step interpolation. Each target uses four true prior LBM fields. Identical colour limits are used for LBM and CNN; the signed error has its own much narrower scale.

Evidence gate

At retained $Re = 105$, the one-step vorticity error is 0.815% for the CNN, 1.053% for matched cubic extrapolation, and 11.009% for persistence. The mean downstream profile errors are 0.804%, 1.581%, and 16.924%, respectively. Thus the CNN beats the strongest declared non-neural baseline; this evidence still supports a one-step claim only.

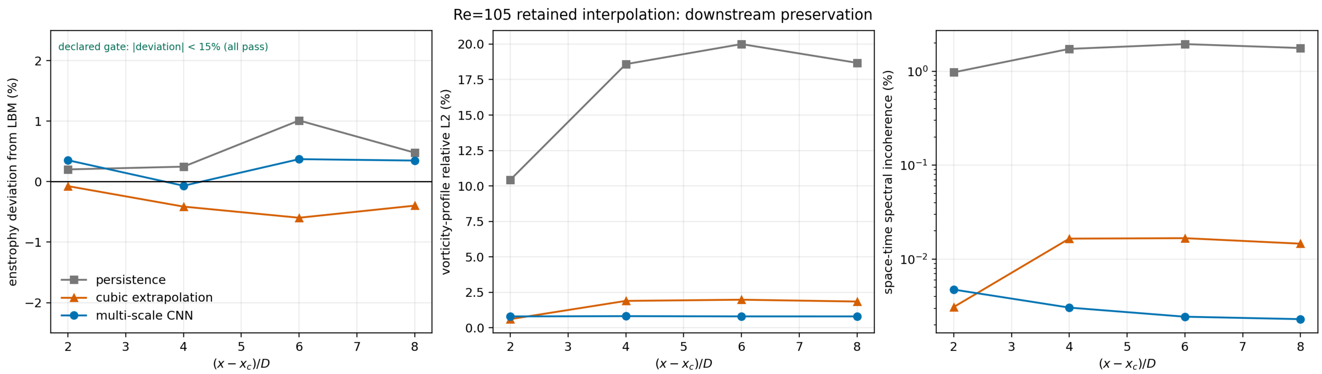


Figure 4: Downstream diagnostics at $2D$, $4D$, $6D$, and $8D$. Enstrophy is shown as percentage deviation rather than on a visually flat 0–1.25 ratio axis. The spectral diagnostic retains complex phase information.

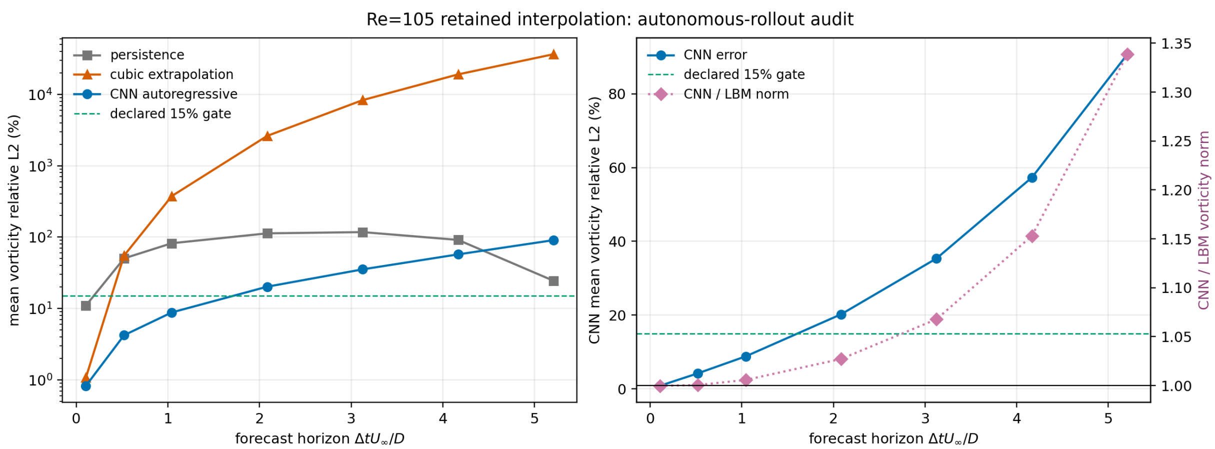


Figure 5: Autonomous-rollout audit from four true initial fields. A failed gate is retained as evidence; it is not converted into a positive claim.

Across six starting phases at retained $Re = 105$, the autoregressive CNN has mean vorticity errors of 8.814% at 10 steps, 20.222% at 20 steps, and 90.724% at 50 steps. It therefore crosses the predeclared 15% gate between 10 and 20 steps. A visually accurate next-frame movie is not evidence of long-horizon stability.

10 Phase-Stable Long-Horizon Correction

The failed CNN rollout does not imply that long-horizon prediction is impossible; it identifies recursive field feedback as the weak link. In the saturated low-Reynolds regime, the cylinder wake is approximately periodic. FlowMLLab therefore adds a separate phase-stable learned Fourier decoder,

$$\widehat{\boldsymbol{q}}(\boldsymbol{x},\phi;\mathrm{Re})=\boldsymbol{a}_0(\boldsymbol{x};\mathrm{Re})+\sum_{k=1}^{K}\left[\boldsymbol{a}_k^s(\boldsymbol{x};\mathrm{Re})\sin(k\phi)+\boldsymbol{a}_k^c(\boldsymbol{x};\mathrm{Re})\cos(k\phi)\right],\qquad \boldsymbol{q}=(u,v,p). \tag{31}$$

The spatial coefficient fields are fitted by least squares to complete development trajectories and interpolated between the two development Reynolds cases that bracket the target. Time evolution is explicit,

$$\phi_n=\phi_0+2\pi\mathrm{St}(\mathrm{Re})n\Delta t^*. \tag{32}$$

Four initial true fields determine ϕ_0 by minimum reconstruction error. Every later field is decoded from the advancing phase, with no future CFD input and no recursive reuse of a predicted field.

Important distinction

This is a learned reduced-order Fourier decoder, not a neural network. The CNN remains the strongest demonstrated one-step model. The phase decoder is the long-horizon correction for a saturated, nearly periodic wake; it does not cover arbitrary transients, new geometries, or unbracketed extrapolation.

10.1 Frozen split and gates

Only $\mathrm{Re}=90,110,120,140$ are used to fit coefficient fields. Harmonic orders $K=2,\ldots,6$ are compared on complete-case $\mathrm{Re}=100$ validation, which selects $K=6$. That choice is frozen before opening the fresh $\mathrm{Re}=95$ test. The former $\mathrm{Re}=105$ case remains a retained historical test. The predeclared long-horizon gates are

$$\epsilon_{\omega,\mathrm{global}}<15\%,\qquad \max_n \epsilon_\omega(n)<15\%,\qquad \frac{|\widehat{\mathrm{St}}-\mathrm{St}_{\mathrm{CFD}}|}{\mathrm{St}_{\mathrm{CFD}}}<2\%. \tag{33}$$

Split	Re	global ω error	worst frame	St error
Validation	100	6.037%	7.959%	0.343%
Fresh test	95	4.281%	5.194%	0.264%
Retained test	105	4.625%	6.240%	0.253%

All entries cover 277 future frames after four true initial fields, and all three gates pass on every split.

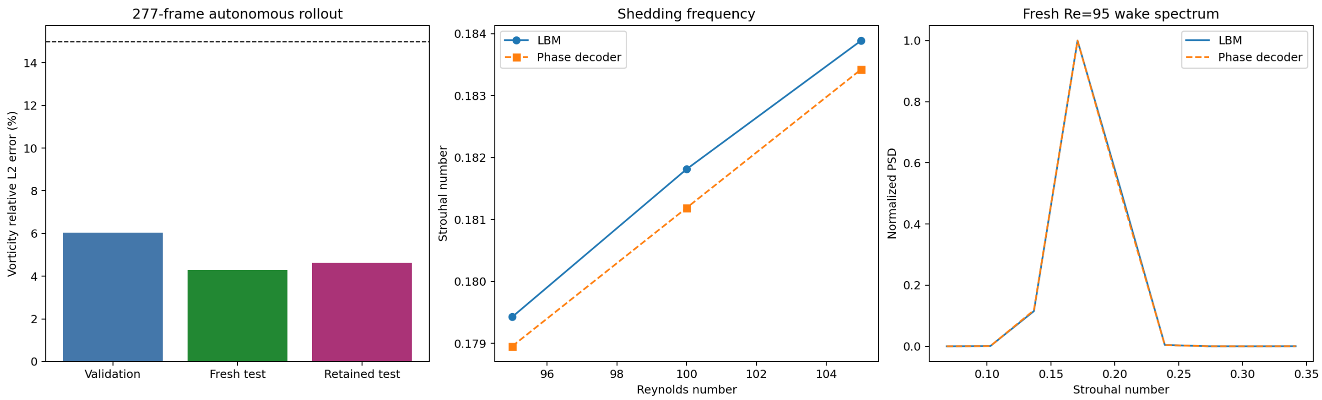


Figure 6: Leakage-controlled long-horizon evidence. Left: global vorticity error against the predeclared gate. Centre: LBM and decoder Strouhal numbers. Right: overlapping fresh-test wake spectra at $\mathrm{Re} = 95$.

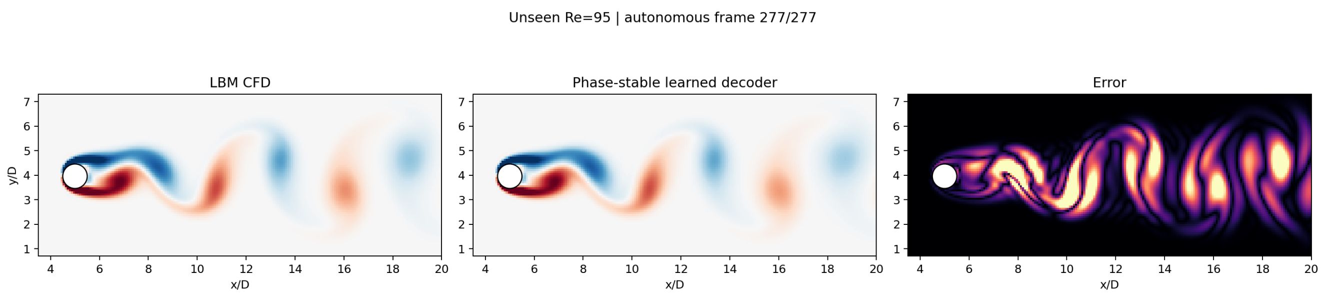


Figure 7: Fresh unseen $\mathrm{Re} = 95$ field comparison after autonomous evolution. The axes use equal physical scale, so the analytical cylinder is displayed as a circle. The error panel has its own colour scale.

At $\mathrm{Re} = 95$, $\mathrm{St}_{\mathrm{CFD}} = 0.179428$ and $\widehat{\mathrm{St}} = 0.178955$. The 0.264% frequency error and the field-error history are evaluated from the same 277-frame window. This is stronger than a single visually selected frame, but it remains validation against the educational LBM labels rather than independent high-fidelity DNS.

11 Connection to the 2019 JFM Study

Connection to Lee & You (JFM, 2019)

Lee and You (2019) developed a deep-learning framework for unsteady circular-cylinder flow prediction using consecutive flow fields, convolutional processing, and recursive future prediction. Their study motivates three principles used here: temporal context must be explicit, evaluation must examine future evolution rather than isolated images, and physical diagnostics are necessary in addition to image similarity.

The present Week-7 exercise is *inspired by*, not a reproduction of, Lee and You. The domain, solver, training set, network capacity, losses, and retained resolution are deliberately smaller for transparent CPU-accessible teaching. Consequently, agreement with one animation cannot be presented as replication of the 2019 result.

The most important comparison is methodological:

Question	Lee & You (2019)	FlowMLLab Week 7
Temporal input	Consecutive flow information	Four consecutive u, v, p fields
Prediction test	Future-field and recursive behavior	Qualified one-step result, failed CNN recursion, and a separate phase-stable rollout
Generalization	Study-specific trained/tested conditions	Complete-case validation plus fresh bracketed $\mathrm{Re} = 95$ test
Physical assessment	Flow evolution beyond visual similarity	L_2, spectra, phase, divergence/no-slip, downstream enstrophy
Purpose	Research framework	Auditable educational extension

Important distinction

A sharp first vortex followed by a diffuse downstream wake usually indicates accumulated phase/-transport error, spectral damping, or an MSE-driven smoothing bias. It is not acceptable to judge the model only near the cylinder.

12 Notebook Workflow and Decision Gates

The companion notebook follows this order:

1. derive D2Q9 equilibrium and verify moment recovery;
2. audit Re, Ma, τ, blockage, and boundary masks;
3. load the five retained regimes and continue the saturated checkpoint;
4. examine C_D, C_L, quality-gated St, time-mean recirculation, and vorticity;
5. reproduce the three-grid table, explain the retained formal failure, and specify the next resolution without treating CFD fields as qualified labels;
6. establish a POD/phase baseline and observe diffusion failure;
7. freeze the four-frame multi-scale CNN using development cases;
8. compare persistence, linear, quadratic, and cubic extrapolation;
9. retain the complete $\mathrm{Re} = 105$ interpolation case and failed CNN rollout;
10. select the phase decoder only on $\mathrm{Re} = 100$ validation;
11. open fresh $\mathrm{Re} = 95$ once and run 277 future fields; and
12. report one-step CNN and long-horizon phase-decoder evidence separately.

Evidence gate

Pass the one-step claim only if the model improves on the best declared polynomial baseline without unacceptable downstream enstrophy loss or space–time spectral/phase failure. Apply a separate predeclared gate to autonomous rollout. If rollout fails, retain that result and restrict the claim; do not describe the one-step animation as rollout.

For the phase decoder, pass the long-horizon claim only when global error, worst-frame error, and Strouhal error all satisfy their frozen gates. Keep the wording restricted to saturated periodic wakes inside the development Reynolds bracket.

13 Questions for Discussion

1. Why can a lower global MSE coexist with a visibly worse vortex street?
2. Which error grows first in rollout: amplitude, phase, wavelength, or wake width?
3. How would a vorticity-gradient, spectral, or transport-aware loss change the bias?
4. Why is an unseen Reynolds case stronger evidence than randomly held-out snapshots?
5. Which LBM error could be learned faithfully by the CNN yet remain physically wrong?
6. Why can a grid-independent result still disagree with published CFD, and which domain or boundary test would you run next?
7. Under constant-Mach grid refinement, why must τ change to preserve Re, and why must that dependent change be reported?
8. Why does explicit phase advancement prevent recursive numerical diffusion but still remain sensitive to a small Strouhal bias?
9. What additional tests would be required before applying this decoder to startup transients, irregular wakes, or a new geometry?

14 References

1. X. He and L.-S. Luo, "Lattice Boltzmann Model for the Incompressible Navier–Stokes Equation," *Journal of Statistical Physics*, 88, 927–944 (1997). doi:10.1023/B:JOSS.0000015179.12689.e4.
2. C. P. Jackson, "A Finite-Element Study of the Onset of Vortex Shedding in Flow Past Variously Shaped Bodies," *Journal of Fluid Mechanics*, 182, 23–45 (1987). doi:10.1017/S0022112087002234.
3. L. Qu, C. Norberg, L. Davidson, S.-H. Peng, and F. Wang, "Quantitative Numerical Analysis of Flow Past a Circular Cylinder at Reynolds Number between 50 and 200," *Journal of Fluids and Structures*, 39, 347–370 (2013). doi:10.1016/j.jfluidstructs.2013.02.007.
4. S. Lee and D. You, "Data-Driven Prediction of Unsteady Flow over a Circular Cylinder Using Deep Learning," *Journal of Fluid Mechanics*, 879, 217–254 (2019). doi:10.1017/jfm.2019.700.
5. M. Bouzidi, M. Firdaouss, and P. Lallemand, "Momentum Transfer of a Boltzmann-Lattice Fluid with Boundaries," *Physics of Fluids*, 13, 3452–3459 (2001). doi:10.1063/1.1399290.
6. I. Ginzburg, F. Verhaeghe, and D. d'Humieres, "Two-Relaxation-Time Lattice Boltzmann Scheme: About Parametrization, Velocity, Pressure and Mixed Boundary Conditions," *Communications in Computational Physics*, 3, 427–478 (2008). publisher page.
7. Q. Zou and X. He, "On Pressure and Velocity Boundary Conditions for the Lattice Boltzmann BGK Model," *Physics of Fluids*, 9, 1591–1598 (1997). doi:10.1063/1.869307.
8. A. J. C. Ladd, "Numerical Simulations of Particulate Suspensions via a Discretized Boltzmann Equation. Part 1," *Journal of Fluid Mechanics*, 271, 285–309 (1994). doi:10.1017/S0022112094001771.
9. R. Mei, D. Yu, W. Shyy, and L.-S. Luo, "Force Evaluation in the Lattice Boltzmann Method Involving Curved Geometry," *Physical Review E*, 65, 041203 (2002). doi:10.1103/PhysRevE.65.041203.
10. S. C. R. Dennis and G.-Z. Chang, "Numerical Solutions for Steady Flow Past a Circular Cylinder at Reynolds Numbers up to 100," *Journal of Fluid Mechanics*, 42, 471–489 (1970). DOI link.

11. S. Sen, S. Mittal, and G. Biswas, "Steady Separated Flow Past a Circular Cylinder at Low Reynolds Numbers," *Journal of Fluid Mechanics*, 620, 89–119 (2009). doi:10.1017/S0022112008004904.

12. C. H. K. Williamson, "Vortex Dynamics in the Cylinder Wake," *Annual Review of Fluid Mechanics*, 28, 477–539 (1996). doi:10.1146/annurev.fl.28.010196.002401.

13. I. B. Celik, U. Ghia, P. J. Roache, C. J. Freitas, H. Coleman, and P. E. Raad, "Procedure for Estimation and Reporting of Uncertainty Due to Discretization in CFD Applications," *Journal of Fluids Engineering*, 130, 078001 (2008). doi:10.1115/1.2960953.